

Optimization

Date _____ Period _____

Solve each optimization problem. You may use the provided box to sketch the problem setup and the provided graph to sketch the function of one variable to be minimized or maximized.

- 1) A supermarket employee wants to construct an open-top box from a 14 by 30 in piece of cardboard. To do this, the employee plans to cut out squares of equal size from the four corners so the four sides can be bent upwards. What size should the squares be in order to create a box with the largest possible volume?

